

STA 130A: Mathematical Statistics: Brief Course

Lecture 6: Discrete Models, Expectation, and Variance

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Fall 2026, UC Davis

Agenda

Last time:

- Counting: permutations and combinations
- Discrete random variables and probability mass function
- Bernoulli and binomial models

Today:

- Complete the common models: geometric and Poisson
- Functions of random variables
- Expectation and variance

Recap: Discrete random variable and PMF

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$

- A random variable assigns a real number to each outcome $\omega \in \Omega$
- The randomness comes from the outcome ω , not from the function X itself
- X is **discrete** if its range is finite or countably infinite

The **probability mass function (PMF)** of X is a function $p_X : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$p_X(x) = P(X = x), \quad x \in \mathbb{R}$$

- $p_X(x) \geq 0$ for all x
- $\sum_x p_X(x) = 1$
- For any set $S \subseteq \mathbb{R}$,

$$P(X \in S) = \sum_{x \in S} p_X(x)$$

Common discrete models: Bernoulli, binomial, geometric, Poisson

Recap: PMFs, Bernoulli, and binomial

For a discrete random variable X ,

$$p_X(x) = P(X = x), \quad \mathcal{X} = \{x : p_X(x) > 0\}, \quad P(X \in S) = \sum_{x \in S \cap \mathcal{X}} p_X(x).$$

Bernoulli: one success/failure trial

$$X \sim \text{Bernoulli}(p) : \quad P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

Binomial¹: successes in n independent trials with common probability p

$$X \sim \text{Binomial}(n, p) : \quad P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n.$$

¹Here $n \geq 1$ and $0 < p < 1$; endpoint cases are deterministic

Geometric random variable

Purpose: Waiting time until the first success in repeated independent Bernoulli(p) trials

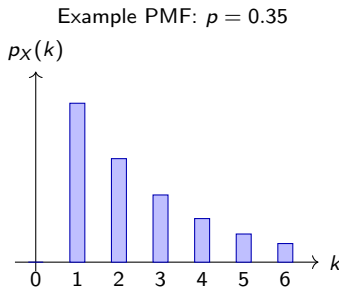
Notation: For $p \in (0, 1]$,

$$X \sim \text{Geometric}(p).$$

PMF:

$$p_X(k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

Support: $\{1, 2, 3, \dots\}$ (for $0 < p < 1$)



- **Interpretation:** $X = k$ means the first success occurs on trial k
- **Example:** Number of tosses until the first Head
- The trials are i.i.d. Bernoulli(p); we stop at the first success

Poisson random variable

Purpose: Number of occurrences in a fixed time or space window

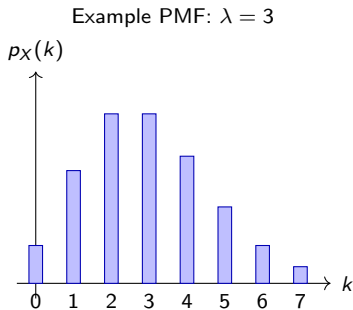
Notation:

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0.$$

PMF:

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Support: $\{0, 1, 2, \dots\}$



- **Interpretation:** $X = k$ means the event occurs k times in the window
- **Example:** Number of emails in an hour; number of typos on a page
- $\text{Binomial}(n, p_n)$ converges to $\text{Poisson}(\lambda)$ as $n \rightarrow \infty$ with $np_n \rightarrow \lambda$

Pop-up quiz

Suppose independent Bernoulli(p) trials are repeated until the first success occurs

Question: Which model best describes the random variable

X = the trial on which the first success occurs?

- A) Bernoulli(p)
- B) Binomial(n, p)
- C) Geometric(p)
- D) Poisson(λ)

Answer: C.

Because X is a waiting time to the first success in repeated independent Bernoulli trials.

Follow-up: If we instead run exactly n trials and count the total number of successes, which model do we get?

Function of a random variable

Given a random variable X , consider its image under a function $g : \mathbb{R} \rightarrow \mathbb{R}$:

$$Y = g(X)$$

- Y is a random variable because $Y = g \circ X$ is a map from Ω to $\mathbb{R}$
- Example: $aX + b$, X^2 , $|X|$, $\log X$ (when $X > 0$), $\dots$

If X is discrete with PMF p_X , then Y is also discrete with PMF p_Y such that

$$p_Y(y) = \sum_{x \in \{x | g(x) = y\}} p_X(x)$$

- Different values of X may map to the same value of $Y = g(X)$

Example

Let X be a discrete RV with $p_X(x) = 1/5$, $\forall x \in \{-2, -1, 0, 1, 2\}$ and let $Y = X^2$

- Y takes only the values 0, 1, 4 $\rightarrow Y$ is discrete
- The PMF of Y satisfies $p_Y(0) = 1/5$, $p_Y(1) = 2/5$, $p_Y(4) = 2/5$

Optional check: Functions of a random variable

Let X satisfy

$$P(X = -1) = \frac{1}{4}, \quad P(X = 0) = \frac{1}{2}, \quad P(X = 1) = \frac{1}{4}.$$

Question: For $Y = X^2$, which statement is correct?

- A) $P(Y = 0) = \frac{1}{4}$
- B) $P(Y = 1) = \frac{1}{2}$
- C) Y can take the values $-1, 0, 1$
- D) Y is not a random variable

Answer: B.

Because both $X = -1$ and $X = 1$ map to $Y = 1$, so

$$P(Y = 1) = P(X = -1) + P(X = 1) = \frac{1}{2}.$$

Follow-up: If g is one-to-one on the values of X , do we still need to sum over several x 's?

Expectation: Definition

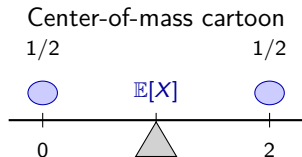
The PMF fully describes the distribution, but often we want a single numerical summary

Definition (Expectation)

The **expectation** (also called the **expected value** or the **mean**) of a random variable X with PMF p_X is

$$\mathbb{E}[X] = \sum_x x p_X(x)$$

- $\mathbb{E}[X]$ is a probability-weighted average of the values of X
- It can be interpreted as the *center of mass* of the PMF
- $\mathbb{E}[X]$ need not be a value that X can actually take



Here $P(X = 0) = P(X = 2) = 1/2$, so $\mathbb{E}[X] = 1$, even though $P(X = 1) = 0$.

Expectation: Example

Example (Two biased coin tosses)

Let X be the number of Heads in two independent tosses, each with

$$P(\text{Head}) = \frac{3}{4}$$

Then

$$P(X = 0) = \frac{1}{16}, \quad P(X = 1) = \frac{6}{16}, \quad P(X = 2) = \frac{9}{16}$$

Therefore,

$$\mathbb{E}[X] = 0 \cdot \frac{1}{16} + 1 \cdot \frac{6}{16} + 2 \cdot \frac{9}{16} = \frac{24}{16} = \frac{3}{2}$$

- We expect 1.5 Heads out of two tosses on average (assuming repeated trials)
- The expected number of Heads is 1.5, even though X can only take the values 0, 1, 2

Variance: Definition

Definition (Variance)

The **variance** of a random variable X is

$$\text{Var}(X) = \mathbb{E} \left[(X - \mathbb{E}[X])^2 \right]$$

- Variance is the average squared distance from the mean
- It is always nonnegative because $(X - \mathbb{E}[X])^2 \geq 0$
- The **standard deviation** of X is defined as $\sigma_X = \sqrt{\text{Var}(X)}$

In principle, we could compute variance by first finding the PMF of $Z = (X - \mathbb{E}[X])^2$

Example

Let X be a discrete RV with $p_X(x) = 1/5$, $\forall x \in \{-2, -1, 0, 1, 2\}$

- $Z = (X - \mathbb{E}[X])^2$ is discrete RV with $p_Z(0) = 1/5$, $p_Z(1) = 2/5$, $p_Z(4) = 2/5$
- Thus, $\text{Var}(X) = \mathbb{E}[Z] = 0 \cdot \frac{1}{5} + 1 \cdot \frac{2}{5} + 4 \cdot \frac{2}{5} = 2$

Expectation of function of random variable

Question: Do we really need the PMF of $(X - \mathbb{E}[X])^2$ in order to compute variance?

Lemma

Let X be a random variable with PMF p_X and $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$$

To verify this, let $Y = g(X)$ and observe

$$\begin{aligned} \mathbb{E}[g(X)] &= \sum_y y p_Y(y) = \sum_y y \sum_{x \in \{x | g(x)=y\}} p_X(x) && \because \text{definition of } p_Y \\ &= \sum_y \sum_{x \in \{x \in \mathbb{R} | g(x)=y\}} g(x) p_X(x) && \because y = g(x) \\ &= \sum_x g(x) p_X(x) \end{aligned}$$

Therefore, $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_x (x - \mathbb{E}[X])^2 p_X(x)$

Expectation of function of random variable: Example

Example

Let X take the values -1 and 1 with probability $1/2$ each, and let $g(x) = x^2$. Then

$$\mathbb{E}[g(X)] = \mathbb{E}[X^2] = 1, \quad g(\mathbb{E}[X]) = g(0) = 0.$$

Example

Let X take the values in $\{1, 2, 3, 4, 5, 6\}$ uniformly at random, and let $g(x) = 1/x$

Then

$$\mathbb{E}[g(X)] = \mathbb{E}\left[\frac{1}{X}\right] = 1 \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{1}{6} + \frac{1}{5} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{147}{360}$$

$$g(\mathbb{E}[X]) = \frac{1}{7/2} = \frac{2}{7}.$$

In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$

Moments

Definition (Moment)

For a positive integer n , the **n -th moment** of a random variable X is $\mathbb{E}[X^n]$

For a discrete random variable,

$$\mathbb{E}[X^n] = \sum_x x^n p_X(x)$$

- The first moment is the mean
- The second *centered* moment is the variance
- Higher (standardized central) moments help describe the shape of a distribution
 - Third order $\rightarrow$ skewness
 - Fourth order $\rightarrow$ kurtosis

In this course, our main focus will be on the mean and the variance

Properties of mean and variance

Lemma

Let X be a random variable and $a, b \in \mathbb{R}$. Then

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\mathbb{E}[aX + b] = \sum_x (ax + b) p_X(x) = a \underbrace{\sum_x x p_X(x)}_{=\mathbb{E}[X]} + b \underbrace{\sum_x p_X(x)}_{=1} = a\mathbb{E}[X] + b$$

$$\begin{aligned} \text{Var}(aX + b) &= \sum_x (ax + b - \mathbb{E}[aX + b])^2 p_X(x) = \sum_x (ax + b - a\mathbb{E}[X] - b)^2 p_X(x) \\ &= a^2 \underbrace{\sum_x (x - \mathbb{E}[X])^2 p_X(x)}_{=\text{Var}(X)} = a^2 \text{Var}(X) \end{aligned}$$

Variance revisited

Lemma

For a discrete random variable X ,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\begin{aligned}\text{Var}(X) &= \sum_x (x - \mathbb{E}[X])^2 p_X(x) \\&= \sum_x (x^2 - 2x \mathbb{E}[X] + (\mathbb{E}[X])^2) p_X(x) \\&= \sum_x x^2 p_X(x) - 2\mathbb{E}[X] \sum_x x p_X(x) + (\mathbb{E}[X])^2 \sum_x p_X(x) \\&= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 \\&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

Pop-up quiz

Let X take the values 0 and 2 with probability $1/2$ each.

Question: Which statement is correct?

- A) $\mathbb{E}[X] = 1$ and $\text{Var}(X) = 0$
- B) $\mathbb{E}[X] = 1$ and $\text{Var}(X) = 1$
- C) $\mathbb{E}[X] = 2$ and $\text{Var}(X) = 1$
- D) $\mathbb{E}[X] = 1$ and $\text{Var}(X) = 2$

Answer: B.

The mean is $0 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 1$, and the average squared distance from 1 is

$$\frac{(0 - 1)^2 + (2 - 1)^2}{2} = 1.$$

Follow-up: Can two random variables have the same mean but different variances?

Example: Bernoulli random variable

Example

If $X \sim \text{Bernoulli}(p)$, then

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

Therefore,

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p,$$

and since $X^2 = X$,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

- If $X = \mathbf{1}_A$ is the indicator of an event A , then

$$\mathbb{E}[X] = P(A), \quad \text{Var}(X) = P(A)(1 - P(A)).$$

Example: Discrete uniform random variable

Example

Let X be the outcome of a fair die roll, so

$$P(X = x) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6.$$

Then

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2},$$

and

$$\mathbb{E}[X^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}.$$

Therefore,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$

Optional example: Derivation of Poisson mean and variance

Example

If $X \sim \text{Poisson}(\lambda)$, then

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^k}{(k-1)!} = \lambda \sum_{j=0}^{\infty} \underbrace{e^{-\lambda} \frac{\lambda^j}{j!}}_{=p_X(j)} = \lambda$$

Also,

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = \lambda^2,$$

so

$$\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = \lambda^2 + \lambda$$

Therefore,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda$$

Application: Decision making with expected values

Expected value is often used to compare decisions with random payoffs

- To maximize long-run average payoff, we typically choose the larger *expected payoff*

Example (Choosing between two lotteries)

Suppose you must choose between

A : receive \$50 for sure, B : receive \$100 with probability 0.6 and \$0 otherwise.

Then

$$\mathbb{E}[\text{payoff under } A] = 50, \quad \mathbb{E}[\text{payoff under } B] = 60.$$

If the goal is to maximize average payoff, you should choose B .

- Expected value is a useful criterion for repeated decisions
- It is not the only consideration though: two options may have the same mean but different risk (measured in variance)

Pop-up quiz

Two lotteries have the same expected payoff:

A : receive \$50 for sure, B : receive \$100 with probability $1/2$ and \$0 otherwise.

Question: Which statement is correct?

- A) Lottery A has a larger expected payoff.
- B) Lottery B has a larger expected payoff.
- C) The two lotteries have the same expected payoff, so expected value alone does not distinguish them.
- D) The two lotteries have the same variance.

Answer: C.

Both have expected payoff 50, but B is riskier. Here, mean alone does not tell the whole story.

Follow-up: What numerical summary might help distinguish the risk of the two options?

Wrap-up

Functions of random variables

- If $Y = g(X)$, then Y is again a random variable
- For a discrete X , compute $p_Y(y)$ by adding the probabilities of all x that satisfy $g(x) = y$

Expectation and variance

- $\mathbb{E}[X] = \sum_x x p_X(x)$ is a probability-weighted average (center of mass)
- $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$ measures spread around the mean
- For any function g ,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$$

- A useful identity:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Suggested reading: [BT08, Ch. 2.1–2.4]

Next lecture: Joint and marginal PMFs; functions of multiple random variables

References



Dimitri Bertsekas and John N Tsitsiklis.

Introduction to probability, volume 1.

Athena Scientific, 2nd edition, 2008.